UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE O Level

MARK SCHEME for the May/June 2006 question paper

5090 BIOLOGY

5090/02 **Paper 2** **maximum raw mark 80**

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Section A

1 (a) *mark awarded only if structure is in a plausible position*

(i) nucleus/cytoplasm/(shown in both cells) ;

(ii) Any 2 from: chloroplast/wall/(cell) sap/membrane ;; **[3]**

(b) (i) photosynthesis ;

manufactures or stores CHO/sugar/glucose/cellulose ; **[2]**

(ii) liver/muscle ;

*storage/cells contain ;

*<u>glycogen</u> ; **[3]**
(**mark separately from liver/muscle mark*)

(iii) muscles largely protein/contain fat ;

skin largely protein ;

animal cells/tissues/skin stores fat ;

fat insulates against heat loss ; **[max 3]**

2 (a) hormones ;

<u>target</u> ; **[2]**

(b) 1. C/blood glucose rises ;

2. E/heart beat increases ; **[2]**

(c) (i) I (or otherwise identified) ;

greatest control over sugar level/smallest fluctuations AW ;

at lowest (blood glucose) level ;

(ii) H ;

greatest fluctuations/little control over sugar levels ; **[max. 4]**

(d) lungs ;

alveoli/air sacs ;

<u>diffusion</u> ;

into <u>capillaries</u> ; **[max 3]**

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3	**(a)**	pollen	;		**[1]**
	(b)	by insect	;		
		grain sticky/rough AW	;		**[2]**
	(c)	fusion AW	;		
		male and female	;		
		gametes/nuclei/sex cells	;		
		fertilisation	;		
		ref. food storage	;		
		mitosis/growth	;		
		embryo development	;		**[max 4]**
	(d)	(seed) dispersal (ignore refs. to wind)	;		**[1]**
4	**(a)**	any 2 from : urination/exhaling or breathing out/faeces/ bleeding or crying or vomiting	;;		**[2]**
	(b)	**(i)** higher when walking (or v.v.)/quoted figures	;		
		more energy/heat released/raises body temperature	;		**[2]**
		(ii) lower when clothed (or v.v.)/quoted figures	;		
		greater humidity next to skin/(v.v.) less skin exposed/ clothes deflect or absorb heat AW	;		**[2]**
		(iii) higher in sun (or v.v.)/quoted figures	;		
		higher temperatures in direct sunlight/higher rate of evaporation	;		**[2]**
	(c)	more energy released/respiration/work done by	;		
		muscles	;		**[2]**
5	**(a)**	**(i)** 105	;		**[1]**
		(ii) genes/alleles (A any given pair of contrasted characters)	;		**[1]**
		(iii) to prevent choice/bias/so results are random	;		**[1]**
	(b)	**(i)** red + **W**	;		**[1]**
		(ii) ref. both cubes and both flowers being the same/heterozygous AW/ the only way to produce both colours of offspring/gives all genetic combinations AW	;		**[1]**

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(c) **(i)** T t + red* ;

(x) tt + yellow* ;

gametes ;

gametes correctly shown (need be once only for tt) ;

genotypes of offspring correctly derived (* A colour tie-up here) ; **[max. 4]**

(ii) 3 x T + 3 x t on one cube + 6 x t on the other ; **[1]**

The maximum for Section A = 50 marks

Section B

6 **(a)** active site ;

of specific shape AW ;

substrate ;

fit/are complementary ;

any ref. enzyme/substrate complex being like lock and key ;

stress on substrate molecule ;

product formed ;

also works in reverse ; **[max 5]**

(b) reaction rate increases ;

similar to key turning more often ;

more energy/faster movement of molecules ;

active site changes shape ;

proteins are denatured by heat AW ;

permanently ;

reaction stops ;

substrate no longer fits active site ;

key no longer fits lock ; **[max 5]**

[Total = 10]

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7 **(a)** named e.g. of bacterial disease ;

named method of administration ;

antibiotics kill only bacteria ;

must continue with course until all bacteria are eliminated ;

named antibiotic ; **[max 3]**

(b) fermenter/vat/large container ;

culture medium ;

addition of organism (fungus or bacterium) ;

controlled temperature ;

provision of oxygen ;

conditions optimum/controlled for maximum production ;

extraction of antibiotic ;

purification ; **[max 7]**

[Total = 10]

8 **E** **(a)** traps/harnesses/absorbs ;

sunlight ;

energy ;

for photosynthesis ;

which makes carbohydrate AW ; **[max 4]**

(b) large surface area ;

for maximum/rapid ;

uptake of water ;

by osmosis/diffusion ;

of ions/salts/minerals ;

by active transport ;

oxygen ;

for root respiration ; **[max 6]**

[Total = 10]

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8	O	**(a)**	absorbs + quickly	;	
			and carries	;	
			oxygen	;	
			as oxyhaemoglobin	;	
			in red blood cells	;	**[max 4]**
		(b)	large surface area	;	
			uptake from ileum/small intestine	;	
			*of amino acids	;	
			*of glucose	;	
			into blood capillaries	;	
			*fats/fatty acids/glycerol	;	
			into lacteals (** allow one for digested foods*)	;	**[max 6]**
					[Total = 10]

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